A concrete 263-bit prime conductor factor

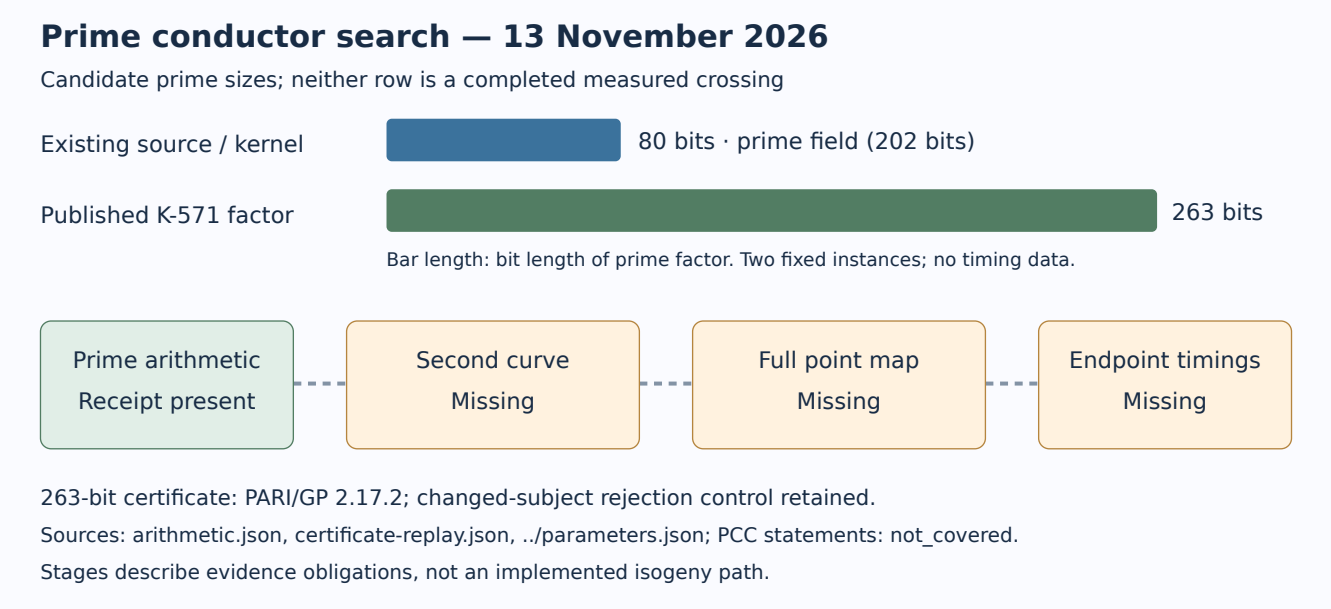
13 November 2026 · Fixed published-instance replay and construction review.

The K-571 example of Koblitz–Koblitz–Menezes has a 263-bit prime factor in its Frobenius conductor. PARI/GP 2.17.2 reconstructed the factor, produced an ECPP certificate and accepted that certificate in a separate replay process. Changing the subject caused rejection. These arithmetic observations are PCC statements, with verdict not_covered; independent review remains required: [pcc:unverified]. Data claim: PCC-20261113-add2c4.

Prime:

9021184135396238924389891945192676814518993645089807769277009849103733654828039

The curve is K-571, $y^2 + xy = x^3 + 1$ over $F_{2^{571}}$. The source paper gives endomorphism conductor 1 and fundamental discriminant −7. The replay's Frobenius conductor is the product of the displayed prime and 3952463, a 22-bit prime. Full trace, conductor and checks are retained in arithmetic.json. This is a published factor, not a newly evaluated crossing.



Acceptance criterion and missing components

The requested minimum remains an 80-bit prime endomorphism-conductor gap between ordinary elliptic curves, crossed by a completely evaluated isogeny, with matched endpoint ECDLP measurements. Both the existing 80-bit source and this 263-bit published factor lack a supplied quotient and full point map. No new ECDLP or scalar-multiplication timings were produced. A full K-571 floor has composite conductor; an exactly prime ratio needs an appropriate intermediate endpoint, also absent here.

Construction assessment

Galbraith–Gilchrist–Robert Algorithm 3 recovers a vertical map from two supplied curves. It does not produce the missing endpoint. Its extra discriminant-divisor mechanism has only 1 and 7 available at our maximal discriminant −7 endpoint. Higher-dimensional evaluation still needs initial map information. Horizontal invertible ideal actions do not resolve general conductor-changing evaluation; The Isogeny Problems lists the ordinary non-invertible-ideal case as Problem 5.2. This is a bounded assessment, not an impossibility theorem. [pcc:unverified]

Replay and retained evidence

Use an empty output directory because GP write appends. Run check-published.gp with GP 2.17.2, a 128 MB stack, and inspect both stderr and the output files; GP can exit zero after an interpreter error. Then run check-certificate.gp in the same directory. Commands are in ../README.md.

- arithmetic.json: exact trace, conductor, two factors and scope.
- prime263.cert: saved PARI primality certificate (2547 bytes).
- certificate-replay.json: subject binding, acceptance and changed-subject control.
- primecert-stack8M.stderr: initial default-stack exhaustion, retained as failure.
- source-review.json and source-hashes.txt: sources, locators and downloaded-byte hashes.

The 128 MB replay completed; no performance claim is inferred from its duration. This is arithmetic verification of a fixed published instance, not a new search or experiment. No quotient, transported generator or endpoint timing is present. No new subgroup workload was introduced for K-571. [pcc:unverified]

Sources

1. Koblitz, Koblitz and Menezes, Section 11.2, Example 4: author PDF.
2. Galbraith, Climbing and descending tall isogeny volcanos, introduction, Section 5, Appendix A: ANTS submission PDF. This copy predates later IACR revisions.
3. Galbraith, Gilchrist and Robert, Algorithm 3, Proposition 5.1: author PDF, dated 2025-08-15.
4. The Isogeny Problems, Problem 5.2.

Literature claim: PCC-20261113-80903b. KB unavailable: no MCP service exposed, repository wrapper exited 127. Fallback: scoped repository evidence and primary sources. ecdb unavailable in the fresh runtime; fixed-instance arithmetic replay is recorded instead. No exhaustive world-record claim is made.

The existing readiness, complete-pipeline and CM-orbit figures have no changed measurement inputs. No canonical benchmark graph or scoreboard row is promoted. The next technical obligation is a complete quotient and non-kernel images for the existing 80-bit source, with total construction and transport costs recorded.